

DHARMESH MOHANRAJ

Santa Barbara, CA · dharmeshmohanraj@ucsb.edu · +1 805-837-5300 · LinkedIn [↗](#) · GitHub [↗](#) · Portfolio [↗](#)

Summary

M.S. ECE @ UCSB. I build dependable systems across **DSP**, **wireless**, **embedded**, and **ML/GenAI**. Recent work includes **Adaptive ISAC-MIMO** for multi-human tracking and a **Wireless Optimization Agent**; more projects below. Certifications include **IBM Agentic AI** and **AWS Cloud Solutions Architect**. Open to roles in DSP, Wireless Communications, Edge-AI, and ML Systems.

Education

- **M.S., Electrical & Computer Engineering** — University of California, Santa Barbara Sep 2024 – Present
Expected Mar 2026. Relevant Coursework: **Multirate DSP**, **Signal Compression**, **Pattern Recognition**
- **B.E., Electronics & Communication Engineering** — Anna University, Chennai Nov 2020 – Jun 2024
Relevant Coursework: **Digital Communication**, **Cryptography**, **Digital Signal Processing**, **IoT Systems**

Technical Skills

- **Programming & Data:** Python, C, MATLAB, SQL ; NumPy, Pandas, OpenCV, PyTorch, TensorFlow, scikit-learn
- **Cloud & IoT:** AWS, GCP, The Things Stack (LoRaWAN) ; MQTT, Modbus, REST APIs
- **GenAI & ML:** LangChain, LangGraph, FAISS/Chroma/Qdrant, Hugging Face, prompt engineering, XGBoost
- **Embedded & Tools:** Arduino, Raspberry Pi, STM32, ESP32 ; Docker, Git/GitHub, Jupyter, Linux

Work Experience

- **Chakra Network Solutions Pvt. Ltd.** — Summer Intern Jun 2024 – Aug 2024
 - Engineered an IoT telemetry pipeline (**MQTT**, **Modbus**, **REST**) for secure, low-latency device communication; standardized payload schemas and retry logic.
 - Analyzed a **7M-row HVAC** dataset (IITM Research Park) to prototype anomaly detection that flagged incipient faults/inefficiencies, reducing unplanned downtime by **about 20–30%**.
 - Built predictive-maintenance workflows that connected IoT data ingestion to trained **supervised ML models**, generating prioritized alerts with threshold-based rankings for operators.
- **Signal Compression Lab, UC Santa Barbara** — Graduate Researcher Jan 2025 – Present
 - Researching **transform-domain TIP** (**temporal interpolation/prediction**) with **superpixel-guided partitioning** for next-gen video codecs.
 - Prototyped **PyTorch** sub-pixel predictors; evaluating **BD-Rate** against motion-compensated baselines on standard sequences.

Selected Projects (Additional projects and full details: Portfolio [↗](#))

- **VIDEX: End-to-End Video Intelligence Pipeline** — AWS + Python May 2025 – Jun 2025
 - Built serverless ingest (**API Gateway** → **Lambda** → **S3**) with CNN-autoencoder compression achieving **0.95 SSIM**; controlled storage costs via **S3 Glacier Deep Archive**.
 - Automated metadata using **AWS Rekognition** (labels/objects/text); implemented semantic search with **FAISS/Chroma**.
 - Benchmarked vector stores: **FAISS 3.21s index / 0.102s avg query** vs **ChromaDB 7.91s / 0.182s**; designed **migration plan** to **OpenSearch + RAG (LangChain/Bedrock)** for cloud-scale retrieval.
- **Adaptive ISAC-MIMO for Multi-Human Tracking** Apr 2025 – Jun 2025
 - Designed a **64-element 2.4 GHz ISAC** base station (**MIMO**, digital **beamforming**); Monte Carlo verified an **8-beam** budget at **1 W EIRP** while meeting comm **SNR ≥ 15** .
 - Implemented **360-degree** sensing (5 s sweep) with **STFT** micro-Doppler, **MUSIC AoA**, and **Kalman** tracking for low-latency steering.
 - Built an activity-weighted **convex optimization** allocator that preserves the comm link and prioritizes high-confidence directions under **Rayleigh** fading.
- **Lightweight AI Agent for Wireless Network Optimization** — LSTM + LangGraph Jul 2025 – Aug 2025
 - Trained an **LSTM** link-state classifier (approx. **98%** on held-out logs).
 - Deployed a **LangGraph** reflection agent with **Self-RAG + LLM** to adapt actions (modulation/FEC, Tx power, band/beam/relay), improving degraded-link recovery by **20%** in simulation compared to a heuristic baseline.

Certifications

- **AWS Generative AI Applications** [↗](#) — Bedrock, prompt engineering, GenAI solution architecture Jul 2025
- **AWS Cloud Solutions Architect** [↗](#) — VPC/IAM/S3/EC2, scalable architectures, cost optimization Jun 2025
- **Introduction to Network Automation** [↗](#) — APIs, Ansible, NetDevOps, model-driven programmability Jul 2025
- **Embedded Software Engineer (C)** [↗](#) — ARM Cortex, Embedded C, UART/SPI/I²C Jun 2025
- **IBM GenAI & Agentic AI** [↗](#) — RAG, vector DBs, LangChain/LangGraph Jul 2025